Depth cleaning and ICP tracking: summary

Kinect-Lite depth denoising, phase 2. Data: TUM fr3 long_office_household (real sensor).

1. The pipeline

depth image -> **clean.py** (gate, median filter, fill holes) -> **icp.py** (point cloud, ICP between neighbouring frames, camera path, error against the true path).

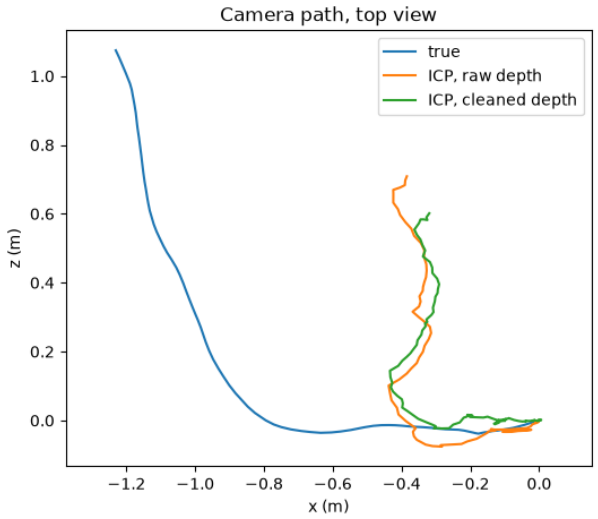
Step	What it does	Why
Gate	Depth closer than 0.4 m becomes a hole.	A Kinect-type sensor cannot measure that close, so such values a
Median filter (9x9)	Each measured pixel takes the median of the measured pixels around it. Holes are ignored.	Removes spiky noise but keeps edges sharp. Median keep
Fill holes	Missing pixels are filled from their surrounding (OpenCV inpainting), limited to small holes.	Gives ICP complete depth. Filled holes are guessed.
ICP	Aligns each frame's point cloud with the previous one, then chains the estimated camera motion.	Standard way to estimate camera motion from depth.
Score (ATE)	Average distance between estimated and true camera positions.	Compares to the true path. Lower is better.

2. Result: which cleaning step helps ICP?

ICP on frames 0-200, every 2nd frame (100 frames). The raw-depth baseline is computed once and saved.

Input to ICP	Error (cm)	vs raw
Raw depth (baseline)	56.45	-
Gate only	56.45	same
Gate + median	51.88	8% better
Full clean (gate + median + fill)	56.51	same

Same test with every frame: raw 57.54, gate + median 52.46, full clean 48.57 cm. So the median helps in both runs (about 9%), while hole filling is inconsistent (best in one run, no gain in the other).





3. Why these choices

- **Median is kept for tracking**: it is the only step with a consistent gain, and it adds no fake depth.
- **Hole filling is optional for tracking**: ICP simply skips missing pixels, and filled values are guesses that can be wrong near object edges. Filling is still useful when a complete depth image is wanted.
- **Gate is kept**: harmless on TUM, and needed for the ICL-NUIM data.
- Earlier on simulated ICL-NUIM data (known true depth), cleaning cut the average depth error from 34.7 mm to 13.0 mm, so the denoising itself works.

4. Limits (read before quoting numbers)

- One stretch of one sequence was tested.
- Every variant is still about 50 cm off, so ICP drifts regardless of cleaning; a 5-10% gain on top of that is small. A better tracker (frame-to-map instead of frame-to-frame) would give a fairer test.
- A longer run (200 frames, every 5th) lost track completely (about 2 m error), so frames must be close together.
- ICL-NUIM ICP is untested: its depth folders are not on disk.